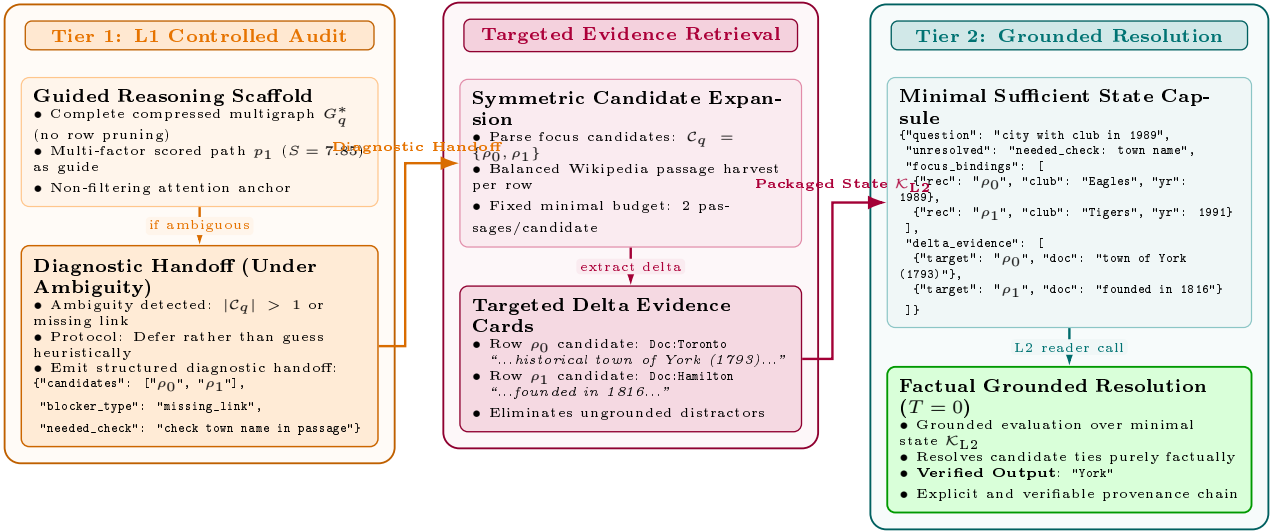

Tier 1: L1 Controlled Audit
Guided Reasoning Scaffold
• Complete compressed multigraph G_q^* (no row pruning)
• Multi-factor scored path p_1 ($S = 7.85$) as guide
• Non-filtering attention anchor
if ambiguous
Diagnostic Handoff (Under Ambiguity)
• Ambiguity detected: $|\mathcal{C}_q| > 1$ or missing link
• Protocol: Defer rather than guess heuristically
• Emit structured diagnostic handoff:
{"candidates": ["ρ_0", "ρ_1"],
"blocker_type": "missing_link",
"needed_check": "check town name in passage"}
Diagnostic Handoff
Targeted Evidence Retrieval
Symmetric Candidate Expansion
• Parse focus candidates: $\mathcal{C}_q = \{\rho_0, \rho_1\}$
• Balanced Wikipedia passage harvest per row
• Fixed minimal budget: 2 passages/candidate
extract delta
Targeted Delta Evidence Cards
• Row ρ_0 candidate: Doc:Toronto
"...historical town of York (1793)..."
• Row ρ_1 candidate: Doc:Hamilton
"...founded in 1816..."
• Eliminates ungrounded distractors
Packaged State $\mathcal{K}_{L2}$
Tier 2: Grounded Resolution
Minimal Sufficient State Capsule
{"question": "city with club in 1989",
"unresolved": "needed_check: town name",
"focus_bindings": [
{"rec": "ρ_0", "club": "Eagles", "yr": 1989},
{"rec": "ρ_1", "club": "Tigers", "yr": 1991}
],
"delta_evidence": [
{"target": "ρ_0", "doc": "town of York (1793)"},
{"target": "ρ_1", "doc": "founded in 1816"}
]}
L2 reader call
Factual Grounded Resolution ($T = 0$)
• Grounded evaluation over minimal state $\mathcal{K}_{L2}$
• Resolves candidate ties purely factually
• Verified Output: "York"
• Explicit and verifiable provenance chain